Streaming Hierarchical Clustering for Concept Mining

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Introduction

streaming data mining
dynamic concept → relational hierarchy
evolving circumstances + user feedback → adaptive reorganization

- Real-time concept mining of high-volume data streams
- Dynamic organization of concepts into a relational hierarchy
- Adaptive reorganization of the concept hierarchy in response to evolving circumstances and user feedback

AFE System (High Speed Content Classification System)

Input -> Word Mapping Table (2^20-1 to 4000) -> counting: c > 15 -> calculating score using known concept (vector)

AFE System

input (hash of words) → WMT (2^20-1 to 4000) → counting > 15 → scoring (using known concept)

Hierarchical Partitioning

Document clustering (bag-of-words)

Top-down divisive approach produces better result than bottom-up

= Divisive clustering

Distribution of hits to expectations: affinity of a document to a given collection

Hardware Design

Use of floating point dashed 8-bit integer system to reduce computational complexity

Pack 4000-dimension (from WMT) array of 8-bit byte into bitmap

Implement 32-dimension vector sum using 32 bit registers (integer calculation)

Calculate multiple dot products using 32-bit register and instructions.

Streaming Hierarchical Partitioning

Assume up-to m documents can be stored on memory (can fast access)

Similarity search using a tree structure, search for the most similar leaf(document) using a greedy approach.

Search for the most similar document and replace the existing document probabilistically based on p calculated through similarity. (This can partially fit concept drift and operates similarly to LRU)